Take-Home Quiz: CRISPR-Cas9 and CRISPR Screen Analysis

Instructions: Answer Questions 1–4. Question 5 is optional. Briefly show your reasoning where appropriate.

---

1. gRNA Length and Genome Specificity

A typical Cas9 guide RNA (gRNA) contains a **20-nt targeting sequence**. The human genome contains approximately **3.1×10^9 base pairs**.

Assuming DNA is composed of four possible nucleotides (A, T, C, and G), what is the **minimum sequence length theoretically required to provide enough possible sequences to uniquely represent the human genome**?

Show how you obtain your answer mathematically.

Hint:

$$4^n \geq 3.1 \times 10^9$$

Based on your calculation, comment briefly on the use of a **20-nt gRNA** for Cas9 targeting.

---

2. Mutations Introduced by Cas9 Gene Editing

Cas9 generates a double-strand break (DSB) in DNA.

What types of mutations can be introduced when the DNA break is repaired by the cell?

In your answer, briefly describe the role of **non-homologous end joining (NHEJ)** and the possible consequences for the coding sequence of a gene.

---

3. CRISPR Gene Editing in Human Disease: Ethical Challenges

Read the following real-world examples of CRISPR/gene-editing applications in humans:

Personalized CRISPR therapy for a child with a rare genetic disease
https://www.chop.edu/news/worlds-first-patient-treated-personalized-crispr-gene-editing-therapy-childrens-hospital

Safety concerns in an early human gene-editing trial
https://www.science.org/content/article/exclusive-death-girl-chinese-gene-editing-trial-was-never-made-public

The CRISPR-edited babies case
https://www.science.org/content/article/chinese-scientist-who-produced-genetically-altered-babies-sentenced-3-years-jail

Write a **short essay of at least 200 words** discussing the major **scientific, clinical, and/or ethical challenges of using gene editing to treat human disease**.

You may use the examples above to support your discussion and are encouraged to provide your own perspective.

---

4. Interpretation of CRISPR Screen Data

The figure below shows two quality-control summaries from a CRISPR screen:

- **sgRNA library coverage** across samples
- **Cumulative distribution function (CDF)** of sgRNA read counts across experimental groups

Study the figure carefully and provide your interpretation of the CRISPR screen data.

Discuss what you can conclude from the figure about:

- sgRNA representation across samples
- differences among the experimental groups
- overall screen/data quality
- any potential technical or biological issues that you notice

Support your conclusions using observations from the figure.

---

5. Optional Hands-on Exercise: CRISPR Screen Data Analysis

Follow the hands-on CRISPR screen data-analysis tutorial provided in class and reproduce the basic workflow on an HPC system or your own computer.

The workflow includes:

```
FASTQ
  ↓
Cutadapt
  ↓
```

```
20-nt sgRNA reads
   ↓
Bowtie1 index
   ↓
Bowtie1 alignment
   ↓
SAM alignment
   ↓
sgRNA count table
   ↓
MAGeCK
   ↓
Differential sgRNA / gene analysis
```

At minimum, practice:

1. Trimming the FASTQ reads using **Cutadapt**
2. Building the sgRNA reference index using **Bowtie1**
3. Aligning the trimmed reads to the sgRNA library
4. Inspecting the alignment output, including the first ~1,000 alignment records
5. Continuing to sgRNA counting and MAGeCK analysis if time permits

Tutorial:
https://github.com/cpdong/share/blob/main/CRISPRseq/Hands-on-CRISPR-Screen-Data-Analysis.md

For this optional exercise, submit the commands you used together with representative output showing that you successfully completed the workflow.